Vanshita Jain

Software Engineer — Backend & AI Tooling
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PROFESSIONAL SUMMARY

Backend engineer applying AI to systematize software delivery — an AI-agent-agnostic development workflow cutting feature time by 50%+, LLM agents that auto-generate data-extraction code, and a RAG-powered semantic search system. Foundation in data engineering (Data Science, IIT Madras) and distributed analytics at scale.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript
Focus: LLM Integration (Claude), RAG, Embeddings, Semantic Search, AI Agents, Distributed Systems, Multi-tenant Architecture, SSO (JWT), RBAC, OLAP
Frameworks: FastAPI, Django, DRF, Flask, Pydantic, SQLAlchemy, Alembic, HTMX, Alpine.js, React
Databases: PostgreSQL, Citus, MySQL, MongoDB, Redis
Infra: Docker, Celery, GCP, Google Pub/Sub, Jenkins, Sentry, Grafana, ClamAV, Cloudflare R2, Playwright, curl_cffi, Gmail API

EXPERIENCE

Tarams Software Technologies | Software Engineer | Remote — January 2025 – Present

JrnyOn* — *Travel Booking & Experiences Platform — *July 2025 – Present*

- Engineered an AI-agent-agnostic dev workflow (shared agent.md spec across Claude, Cursor, Copilot, opencode) automating requirements-to-deployment with 100% diff coverage; cut feature development time by 50%+.
- Built analytics pipeline ingesting ~100K events/day (Mixpanel → Citus OLAP); <200 ms p95 APIs powering 10+ KPI dashboards.
- Architected multi-tenant IAM (subdomain routing, SSO with JWT rotation, per-tenant RBAC) onboarding 3 B2B partners (~20K users) with 0 data leakage and <5 ms auth overhead.
- Engineered a secure image-upload pipeline (ClamAV scan → metadata sanitization → encrypted storage on Cloudflare R2).

Outset (now Button)* — *Retail & Resale Product Enrichment — *January 2025 – July 2025*

- Engineered an AI agent generating per-retailer email parsers from HTML, replacing manual extraction code; cut retailer onboarding time by 50%.
- Designed an Attribute Normalization Engine with weighted synonym mapping; improved classification accuracy from 61% → 91%, eliminating the manual review bottleneck.
- Built an LLM-classified email discovery layer (Gmail search, subject normalization, confidence scoring, Django Admin review), increasing product detection by 15%.

Suraasa | Remote

Software Development Engineer — *July 2023 – January 2025*

- Drove P95 query latency ~1s → ~400 ms for 20K+ concurrent users via composite indexing, SQL refactoring, and Redis caching — zero incidents under full olympiad load.
- Co-owned the backend of Suraasa Jobs — RBAC, listing/application APIs, recruiter-driven school–teacher matching; shipped to 20K+ users.

Software Development Intern — *April 2023 – June 2023*

- Built a mini learning platform from scratch — REST APIs, authentication, and a database schema designed end-to-end.
- Refactored core modules for performance, achieving ~15% faster execution and reducing bug recurrence by ~20%.

PROJECTS

- Job Aggregator & Analyzer — Built a full-stack job aggregator (FastAPI, React, Celery) scraping 3 platforms, with RAG semantic job search (local MiniLM embeddings, cosine retrieval, LLM-extracted negation/location filters, grounded multi-provider LLM answers); JWT auth, Redis rate limiting, 63 pytest tests.
- ACME Salary Management — Built a full-stack HR compensation tool (FastAPI, PostgreSQL, React/TypeScript) designed for 1M employees — DB-trigger-enforced append-only pay history, multi-currency analytics, keyset pagination, pg_trgm search; 579 backend tests at 93% coverage.

EDUCATION

Diploma in Programming | IIT Madras — 2024
B.Sc. (Hons.) Computer Science | Hansraj College, University of Delhi — 2023

CERTIFICATIONS

SystemsExpert | AlgoExpert — 2023
Foundation Level in Data Science and Applications | IIT Madras — 2022